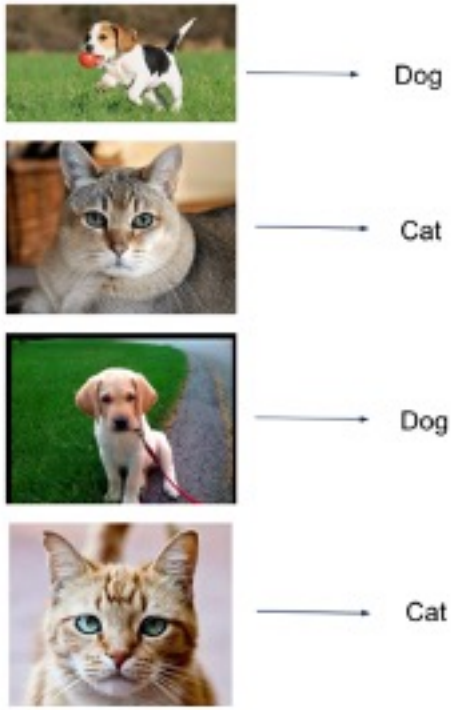
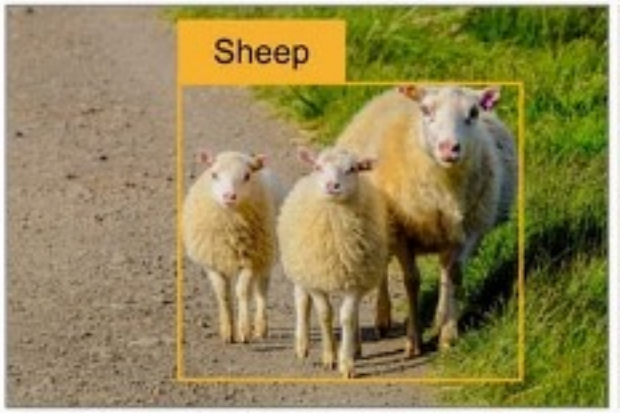
Key Tasks in Computer Vision

Image Classification



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Classification and Localization

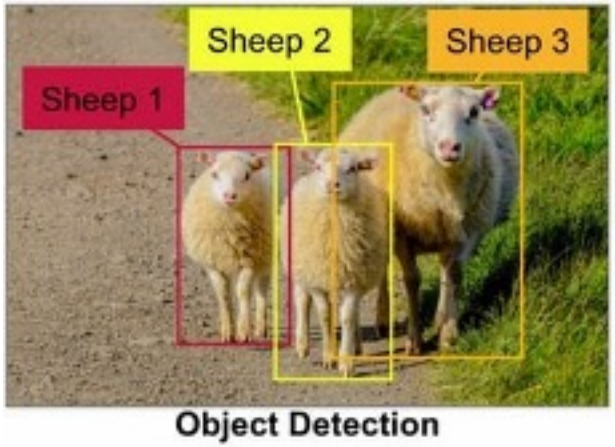


Classification + Localization

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Image Source

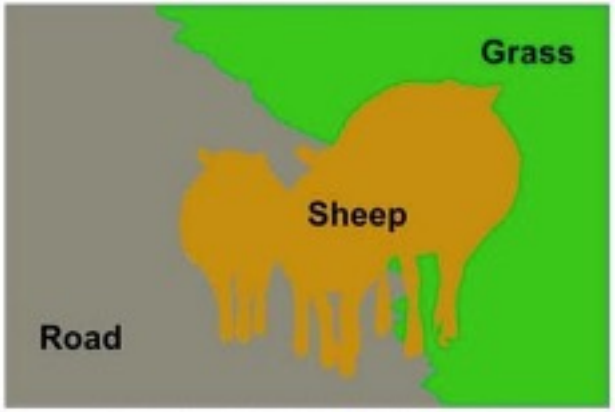
Object Detection



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Image Source

Semantic Segmentation

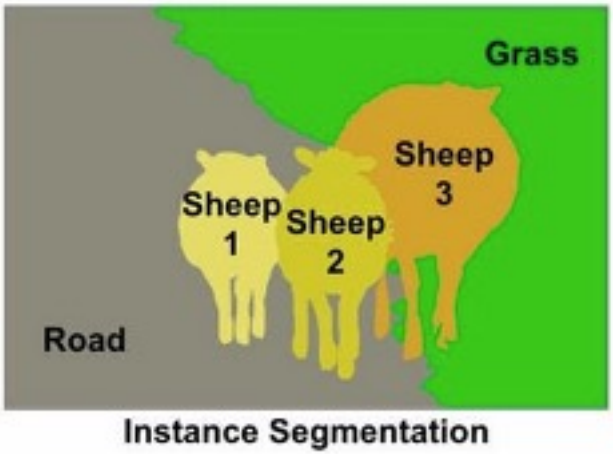


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Every pixel needs to be classified into one of N categories

Image Source

Instance Segmentation



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Every pixel needs to be classified into one of N categories and

Different instances (e.g., Sheep 1, Sheep 2, Sheep 3) of the same category (e.g., Sheep) need to be identified

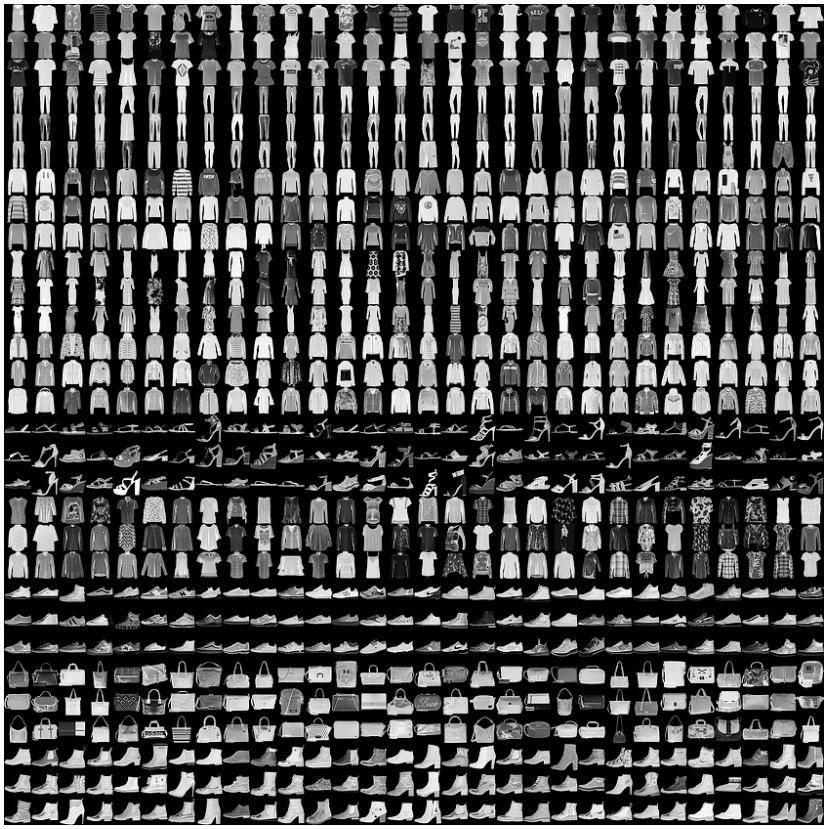
Image Source

Image Classification

Motivating application: Fashion MNIST

The fashion-mnist dataset consists of 70,000 images of clothing items across 10 categories.

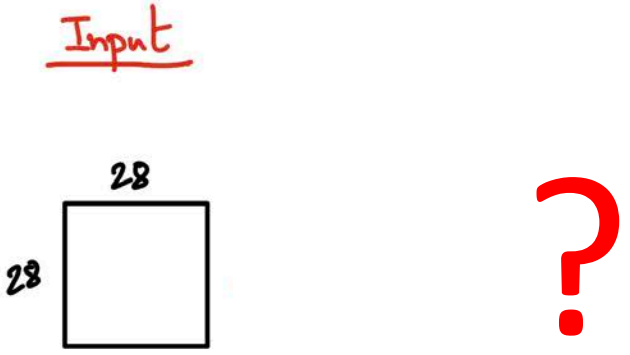
We will build a deep learning network from scratch to classify clothing into these 10 categories with over 90% accuracy!



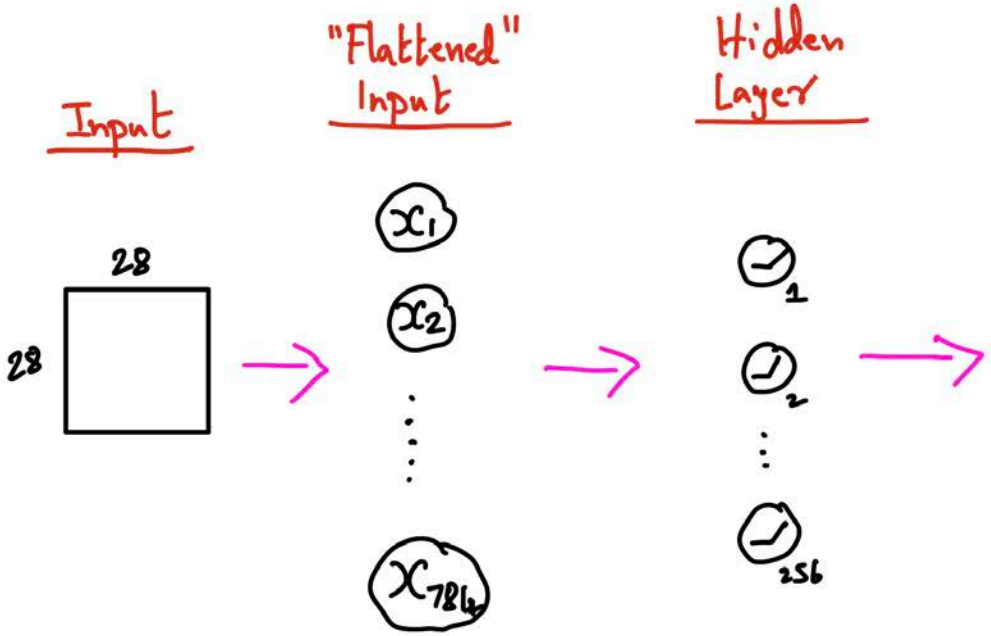
Sample of the Fashion MNIST images by Yuzamei.
Source: Wikimedia Commons. License: CC BY-SA.

Source: https://www.kaggle.com/datasets/zalando-research/fashionmnist

A simple NN to classify grayscale clothing images



A simple NN to classify grayscale clothing images

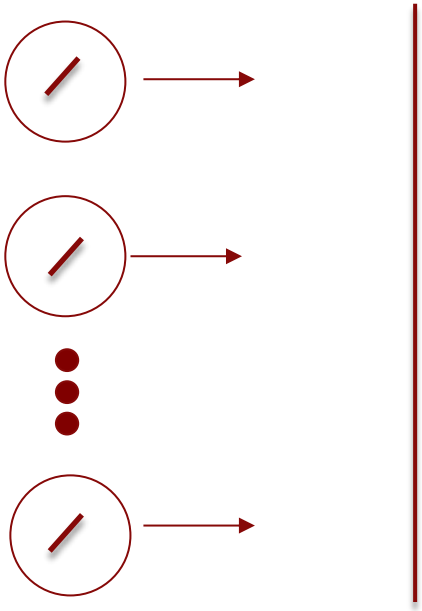


How do we represent a 10-way output?

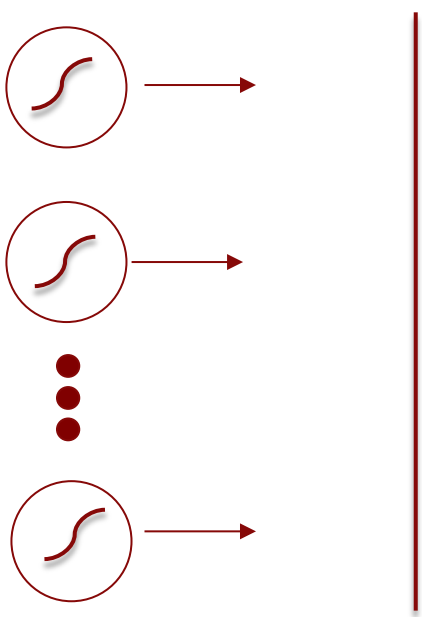
Multi-Class Classification

Suppose the output variable is categorical with 10 levels

We know how to output 10 numbers



We know how to output 10 probabilities



How do we output 10 probabilities that sum to 1.0?

The Softmax Layer

softmax takes in *n* arbitrary numbers and converts them to *n* probabilities

